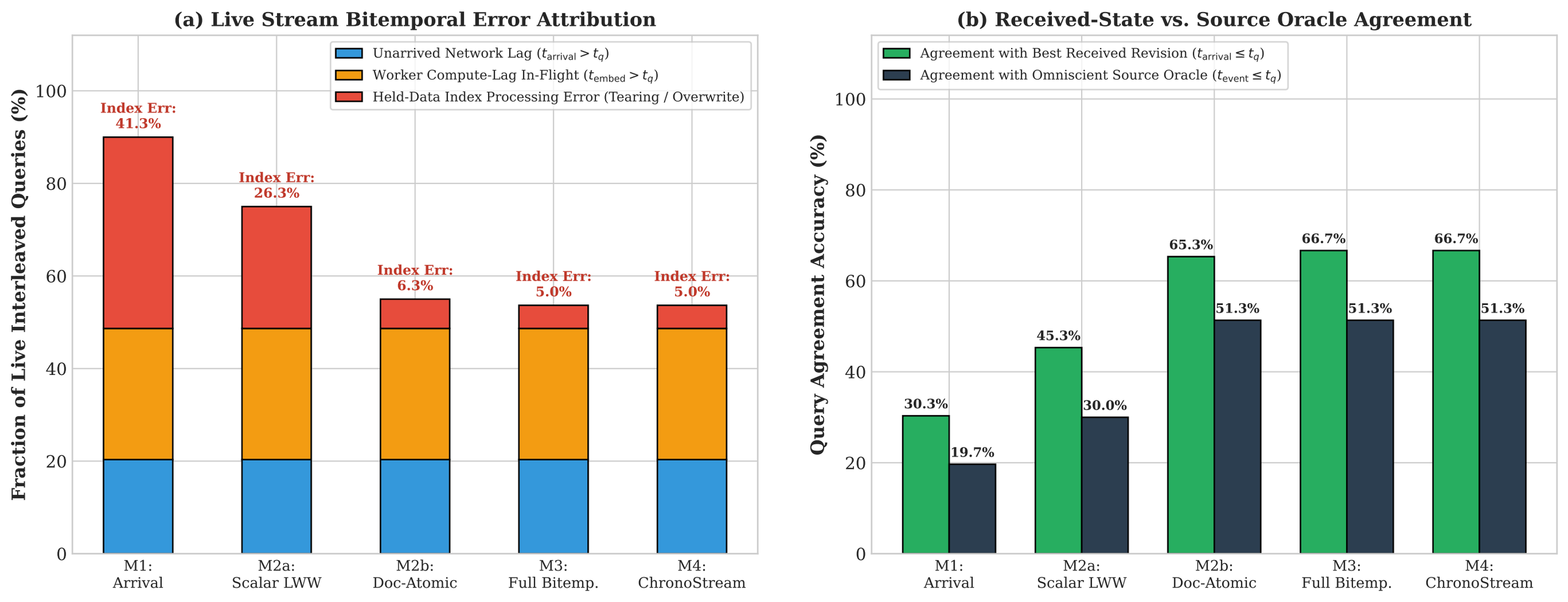
(a) Live Stream Bitemporal Error Attribution
Unarrived Network Lag ($t_{arrival} > t_q$)
Worker Compute-Lag In-Flight ($t_{embed} > t_q$)
Held-Data Index Processing Error (Tearing / Overwrite)
Fraction of Live Interleaved Queries (%)
0
20
40
60
80
100
Index Err: 41.3%
Index Err: 26.3%
Index Err: 6.3%
Index Err: 5.0%
Index Err: 5.0%
M1: Arrival
M2a: Scalar LWW
M2b: Doc-Atomic
M3: Full Bitemp.
M4: ChronoStream
(b) Received-State vs. Source Oracle Agreement
Agreement with Best Received Revision ($t_{arrival} \le t_q$)
Agreement with Omniscient Source Oracle ($t_{event} \le t_q$)
Query Agreement Accuracy (%)
0
20
40
60
80
100
30.3%
19.7%
45.3%
30.0%
65.3%
51.3%
66.7%
51.3%
66.7%
51.3%
M1: Arrival
M2a: Scalar LWW
M2b: Doc-Atomic
M3: Full Bitemp.
M4: ChronoStream